United States Patent Application Publication	20250281188
Kind Code	A1
Publication Date	September 11, 2025
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SYSTEM AND TECHNIQUE FOR METATARSAL REALIGNMENT WITH REDUCED INCISION LENGTH

Abstract

A metatarsal correction procedure may be performed so as to minimize the length of the surgical incision created during the procedure. A variety of different instruments and/or techniques and facilitate the minimal incision procedure. In some examples, a technique involves making a comparatively small surgical incision to access a tarsometatarsal joint. A metatarsal and a cuneiform separated by the tarsometatarsal joint can be prepared. In addition, the metatarsal can be moved relative to the cuneiform using a bone positioning device having a metatarsal engagement member positioned in contact with the skin of the patient on a medial side of the metatarsal. After moving the metatarsal, a fixation device can be installed across the tarsometatarsal joint to cause fusion. In some examples, a plate holder is attached to a U-shaped bone plate to facilitate plating across the joint through the comparatively small incision.

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Family ID: **1000008626647**

Appl. No.: **19/218597**

Filed: **May 26, 2025**

Related U.S. Application Data

parent US continuation 17674447 20220217 parent-grant-document US 12310603 child US 19218597

Publication Classification

Int. Cl.: **A61B17/17** (20060101); **A61B17/88** (20060101)

U.S. Cl.:

CPC **A61B17/1775** (20161101); **A61B17/1728** (20130101); **A61B17/88** (20130101);

Background/Summary

RELATED MATTERS [0001] This application is a continuation of U.S. patent application Ser. No. 17/674,447, filed Feb. 17, 2022, which claims the benefit of U.S. Provisional Patent Application No. 63/151,041, filed Feb. 18, 2021, the entire contents of each of these applications are hereby incorporated by reference.

TECHNICAL FIELD
[0002] This disclosure relates generally to devices and techniques for repositioning bones and, more particularly, to devices and techniques for repositioning bones in the foot.
BACKGROUND
[0003] Bones within the human body, such as bones in the foot, may be anatomically misaligned. For example, one common type of bone deformity is hallux valgus, which is a progressive foot deformity in which the first metatarsophalangeal joint is affected and is often accompanied by significant functional disability and foot pain. The metatarsophalangeal joint is laterally deviated, resulting in an abduction of the first metatarsal while the phalanges adduct. This often leads to development of soft tissue and a bony prominence on the medial side of the foot, which is called a bunion.
[0004] Surgical intervention may be used to correct a bunion deformity. A variety of different surgical procedures exist to correct bunion deformities and may involve removing the abnormal bony enlargement on the first metatarsal and/or attempting to realign the first metatarsal relative to the adjacent metatarsal. Surgical instruments that can facilitate efficient, accurate, and reproducible clinical results are useful for practitioners performing bone realignment techniques.
[0005] For patients, surgical intervention requires making one or more incisions through the patient's skin to access the underlying bones to perform a corrective procedure. A longer incision provides the surgeon with greater access to perform the procedure. However, a longer incision results in a longer scar for the patient after healing, which can be cosmetically undesirable. For this reason, the patient may prefer a shorter incision and/or fewer incisions. This can be challenging for the surgeon because it limits access for performing the corrective procedure.
SUMMARY
[0006] In general, this disclosure is directed to devices, systems, and techniques for performing a metatarsal realignment while minimizing the length and/or number of the incision(s) needed to access the bones for performing the procedure. As a result, a patient undergoing the procedure may have a shorter, less visible incision and resulting scar line, providing healing and/or cosmetic benefits as compared to a procedure performed using a longer incision and/or more incisions.
[0007] In some implementations, a clinician surgically accesses a tarsometatarsal (“TMT”) joint defined between a metatarsal and an opposed cuneiform. The TMT joint may be the first TMT joint between the first metatarsal and medial cuneiform or a lesser TMT joint between a lesser metatarsal and opposed cuneiform. In either case, the clinician can make a comparatively small incision to

access the TMT joint. Once accessed, the clinician can prepare the end face of the metatarsal and the end face of the opposed cuneiform. Example preparation steps may include reaming, cutting, rongeuring, curetting, burring, fenstrating and/or other similar techniques for exposing subchondral bone and/or establishing bleeding bone faces to promote fusion following rejoining of the metatarsal and the cuneiform.

[0008] Either before or after preparation of one or both end faces, the metatarsal is realigned within one or more planes in three-dimensional space. In one example, the clinician engages a bone positioner (also referred to as a bone positioning device) with the metatarsal and a bone other than the metatarsal. The bone positioner can then be actuated to move the metatarsal in one or more planes for realignment. The bone positioner may have a metatarsal engagement member positionable in contact with an external surface of the skin overlying the metatarsal. In other words, instead of making a separate incision to place the bone positioner in contact with the metatarsal bone, the bone positioner can engage the metatarsal bone through the skin to avoid the need for an additional incision.

[0009] In some examples, the bone positioner includes a tip that can be positioned on a lateral side of a metatarsal other than the metatarsal being moved. For example, where the metatarsal being moved is a first metatarsal, a small stab incision may be made on a lateral side of a metatarsal other than the metatarsal being moved, such as between the second metatarsal and the third metatarsal. In these examples, the tip of the bone positioner can be placed in contact with the lateral side of the second metatarsal. The bone positioning may include a mechanism that moves the metatarsal engagement member and the tip towards each other, causing the metatarsal to move in one or more planes to help correct a misalignment of the metatarsal. Additionally or alternatively, the clinician may grasp a pin inserted into the metatarsal to move the metatarsal in one or more planes.

[0010] After moving the metatarsal relative to the opposed cuneiform and/or an adjacent metatarsal, the clinician can perform a variety of steps to complete the surgical procedure. In some examples, the clinician installs a compressor and utilizes the compressor to compress the prepared end face of the metatarsal together with the prepared end face of the cuneiform. Additionally or alternatively, the clinician may temporarily fixate a moved position of the metatarsal relative to the cuneiform. For example, the clinician can insert one or more fixation wires through the metatarsal into one or more adjacent bones (e.g., an adjacent metatarsal, across the TMT joint into the opposed cuneiform).

[0011] In either case, clinician may cause the metatarsal to fuse to the cuneiform. For example, the clinician may install one or more fixation devices across the TMT joint to fixate the position of the metatarsal relative to the cuneiform for subsequent healing and fusion. In some implementations, the clinician installs one more bone plates across the TMT joint. For example, the clinician may utilize a generally U-shaped bone plate have a base and two outwardly extending legs. The clinician may position the base across the TMT joint with the legs extending upwardly in a dorsal direction relative to the base. This type of plate configuration can position one or more fixation holes of the bone plate at a location accessible by the clinician through the comparatively small incision. By contrast, when utilizing a straight plate or other plate configuration, one or more fixation holes of the bone plate may be located subdermally (after inserting the bone place through the incision) in a way that can make it challenging for the clinician to install screws through the fixation holes.

[0012] In some implementations, the clinician may utilize a plate holder to help manipulate and/or hold the bone plate within the incision. The plate holder may be an instrument configured to releasably grasp an external surface of the plate and/or be inserted into a fixation hole of the plate, such as a drill guide inserted through the fixation hole. Using the plate holder, the clinician may be able to controllably and precisely position the bone plate and/or hold the bone plate during installation of one or more screws in the small incision created to access the TMT joint space. This can help address space constrains caused by the comparatively small length incision, which can

make it difficult for the clinician to properly position the bone plate using their fingers along.
[0013] In one example, a method of performing a minimal incision metatarsal correction procedure is described. The method incudes preparing an end of a metatarsal and preparing an end of a cuneiform separated from the metatarsal by a tarsometatarsal joint. The method also involves moving the metatarsal relative to the cuneiform using a bone positioning device that includes a metatarsal engagement member and a tip. The example specifies that the metatarsal engagement member is positioned in contact with a skin of the patient covering a medial side of the metatarsal and that the tip is positioned on a lateral side of a different metatarsal. The method also includes, after moving the metatarsal relative to the cuneiform, causing the metatarsal to fuse to the cuneiform.
[0014] The details of one or more examples are set forth in the accompanying drawings and the description below. Other features, objects, and advantages will be apparent from the description and drawings, and from the claims.

Description

BRIEF DESCRIPTION OF DRAWINGS
[0015] FIGS. **1**A and **1**B are front views of a foot showing a normal first metatarsal position and an example frontal plane rotational misalignment position, respectively.
[0016] FIGS. **2**A and **2**B are top views of a foot showing a normal first metatarsal position and an example transverse plane misalignment position, respectively.
[0017] FIGS. **3**A and **3**B are side views of a foot showing a normal first metatarsal position and an example sagittal plane misalignment position, respectively.
[0018] FIG. **4** is a flow diagram illustrating an example technique for performing a minimally invasive surgical procedure to correct a position of a metatarsal relative to an adjacent bone.
[0019] FIG. **5** is a perspective view of a foot showing example surgical access to a TMT joint.
[0020] FIG. **6** is an image of one example configuration of an incision guide that can be used to help designate the location of a TMT joint prior to making an incision.
[0021] FIG. **7** is a perspective view of a foot showing the incision guide of FIG. **6** positioned over a metatarsal with a radio-identifiable marking line.
[0022] FIG. **8** illustrates one example bone preparation guide that may be used as part of a minimally invasive procedure.
[0023] FIG. **9** illustrates example instrumentation that may be used with the example bone preparation guide of FIG. **8**.
[0024] FIG. **10** is a perspective view of an example foot in which a bi-planar instrument is positioned in adjacent joint spaces.
[0025] FIG. **11** shows a side perspective view of an example bone positioner that can be used to move a metatarsal relative to an adjacent bone.
[0026] FIG. **12** is an illustration showing an example arrangement of the bone positioning device of FIG. **11** in which a metatarsal engagement member is positioned in contact with the external surface of the skin of the patient covering a side of a metatarsal.
[0027] FIG. **13** is a perspective view of a foot showing a seeker and a fulcrum inserted between adjacent joint spaces and a bone positioner engaged with a first metatarsal.
[0028] FIG. **14** is a perspective view of the foot from FIG. **13** further illustrated with a bone preparation guide installed on the seeker.
[0029] FIG. **15** is a further illustration of the surgical instrument arrangement of FIG. **14** showing the example location of an incision and skin of the patient relative to the surgical instruments.
[0030] FIG. **16** is a perspective view of a foot illustrating an example compressing instrument that may be used to compress the prepared end faces of the metatarsal and medial cuneiform together.

[0031] FIG. **17** is illustration of an example bone plate that can be used to help permanently fixate a moved position of a metatarsal relative to an opposed cuneiform.

[0032] FIG. **18** is a perspective view of a foot showing an example insertion location of the bone plate of FIG. **17**.

[0033] FIGS. **19-21** are illustrations of example plate holder configurations that may be used to manipulate a bone plate during a minimal incision procedure.

[0034] FIG. **22** is an illustration of a foot showing an example plate holder manipulating a plate for placement across a TMT joint.

DETAILED DESCRIPTION

[0035] This disclosure generally relates to devices, systems, and techniques for performing a minimal incision bone realignment procedure. In an exemplary application, the devices and techniques can be used during a surgical procedure performed on one or more bones, such as a bone alignment, osteotomy, fusion procedure, fracture repair, and/or other procedures where one or more bones are to be set in a desired position. Such a procedure can be performed, for example, on bones (e.g., adjacent bones separated by a joint or different portions of a single bone) in the foot or hand, where bones are relatively small compared to bones in other parts of the human anatomy. In one example, a procedure utilizing an embodiments of the disclosure can be performed to correct an alignment between a metatarsal (e.g. a first metatarsal) and a cuneiform (e.g., a medial cuneiform), such as a bunion correction. An example of such a procedure is a lapidus procedure. In another example, the procedure can be performed by modifying an alignment of a metatarsal (e.g. a first metatarsal). An example of such a procedure is a basilar metatarsal osteotomy procedure.

[0036] Preparation and fusion of two opposed bone portions, such as a metatarsal and cuneiform, may be performed according to the disclosure for a variety of clinical reasons and indications. Preparation and fusion of a metatarsal and cuneiform at the TMT joint may be performed to treat hallux valgus and/or other bone and/or joint conditions.

[0037] Hallux valgus, also referred to as hallux abducto valgus, is a complex progressive condition that is characterized by lateral deviation (valgus, abduction) of the hallux and medial deviation of the first metatarsophalangeal joint. Hallux valgus typically results in a progressive increase in the hallux abductus angle, the angle between the long axes of the first metatarsal and proximal phalanx in the transverse plane. An increase in the hallux abductus angle may tend to laterally displace the plantar aponeurosis and tendons of the intrinsic and extrinsic muscles that cross over the first metatarsophalangeal joint from the metatarsal to the hallux. Consequently, the sesamoid bones may also be displaced, e.g., laterally relative to the first metatarsophalangeal joint, resulting in subluxation of the joints between the sesamoid bones and the head of the first metatarsal. This can increase the pressure between the medial sesamoid and the crista of the first metatarsal head.

[0038] While techniques and devices are described herein particularly in connection with the first metatarsal and medial cuneiform of the foot, the techniques and devices may be used on other adjacent bones separated by a joint in the hand or foot. For example, the techniques and devices may be performed on a different metatarsal (e.g., second, third, fourth, or fifth metatarsal) and its opposed cuneiform/cuboid. Accordingly, reference to a first metatarsal and medial cuneiform herein may be replaced with other bone pairs without departing from the scope of the disclosure.

[0039] To further understand example techniques of the disclosure, the anatomy of the foot will first be described with respect to FIGS. **1-3** along with example misalignments that may occur and be corrected according to the present disclosure. A bone misalignment may be caused by hallux valgus (bunion), a natural growth deformity, and/or other condition.

[0040] FIGS. **1**A and **1**B are front views of foot **200** showing a normal first metatarsal position and an example frontal plane rotational misalignment position, respectively. FIGS. **2**A and **2**B are top views of foot **200** showing a normal first metatarsal position and an example transverse plane misalignment position, respectively. FIGS. **3**A and **3**B are side views of foot **200** showing a normal first metatarsal position and an example sagittal plane misalignment position, respectively. While

FIGS. **1**B, **2**B, and **3**B show each respective planar misalignment in isolation, in practice, a metatarsal may be misaligned in any two of the three planes or even all three planes. Accordingly, it should be appreciated that the depiction of a single plane misalignment in each of FIGS. **1**B, **2**B, and **3**B is for purposes of illustration and a metatarsal may be misaligned in multiple planes that is desirably corrected. Further, a bone condition treated according to the disclosure may not present any of the example misalignments described with respect to FIGS. **1**B, **2**B, and **3**B, and it should be appreciated that the disclosure is not limited in this respect.

[0041] With reference to FIGS. **1**A and **2**A, foot **200** is composed of multiple bones including a first metatarsal **210**, a second metatarsal **212**, a third metatarsal **214**, a fourth metatarsal **216**, and a fifth metatarsal **218**. The metatarsals are connected distally to phalanges **220** and, more particularly, each to a respective proximal phalanx. The first metatarsal **210** is connected proximally to a medial cuneiform **222**, while the second metatarsal **212** is connected proximally to an intermediate cuneiform **224** and the third metatarsal is connected proximally to lateral cuneiform **226**. The fourth and fifth metatarsals **216**, **218** are connected proximally to the cuboid bone **228**. The joint **230** between a metatarsal and respective cuneiform (e.g., first metatarsal **210** and medial cuneiform **222**) is referred to as the tarsometatarsal ("TMT") joint. The joint **232** between a metatarsal and respective proximal phalanx is referred to as a metatarsophalangeal joint. The angle **234** between adjacent metatarsals (e.g., first metatarsal **210** and second metatarsal **212**) is referred to as the intermetatarsal angle ("IMA").

[0042] As noted, FIG. **1**A is a frontal plane view of foot **200** showing a typical position for first metatarsal **210**. The frontal plane, which is also known as the coronal plane, is generally considered any vertical plane that divides the body into anterior and posterior sections. On foot **200**, the frontal plane is a plane that extends vertically and is perpendicular to an axis extending proximally to distally along the length of the foot. FIG. **1**A shows first metatarsal **210** in a typical rotational position in the frontal plane. FIG. **1**B shows first metatarsal **210** with a frontal plane rotational deformity characterized by a rotational angle **236** relative to ground, as indicated by line **238**.

[0043] FIG. **2**A is a top view of foot **200** showing a typical position of first metatarsal **210** in the transverse plane. The transverse plane, which is also known as the horizontal plane, axial plane, or transaxial plane, is considered any plane that divides the body into superior and inferior parts. On foot **200**, the transverse plane is a plane that extends horizontally and is perpendicular to an axis extending dorsally to plantarly (top to bottom) across the foot. FIG. **2**A shows first metatarsal **210** with a typical IMA **234** in the transverse plane. FIG. **2**B shows first metatarsal **210** with a transverse plane rotational deformity characterized by a greater IMA caused by the distal end of first metatarsal **210** being pivoted medially relative to the second metatarsal **212**.

[0044] FIG. **3**A is a side view of foot **200** showing a typical position of first metatarsal **210** in the sagittal plane. The sagittal plane is a plane parallel to the sagittal suture which divides the body into right and left halves. On foot **200**, the sagittal plane is a plane that extends vertically and is perpendicular to an axis extending proximally to distally along the length of the foot. FIG. **3**A shows first metatarsal **210** with a typical rotational position in the sagittal plane. FIG. **3**B shows first metatarsal **210** with a sagittal plane rotational deformity characterized by a rotational angle **240** relative to ground, as indicated by line **238**.

[0045] Surgical techniques and instruments according to the disclosure can be useful to correct a misalignment of one or more bones, such as the metatarsal and opposed cuneiform, and/or promote fusion of the metatarsal and cuneiform across the TMT joint. In some applications, the technique involves forming two incisions: a comparatively small incision to access the TMT joint and a second stab incision between two metatarsals (e.g., between a second metatarsal and a third metatarsal). The end of the metatarsal and the end of the cuneiform can be prepared. For example, the clinician can insert a bone preparation instrument through the comparatively small incision to prepare the end face of one or both bones.

[0046] Before or after preparing one or both ends, a bone positioner having a metatarsal engagement member and tip can be installed to help facilitate realignment of the metatarsal relative to the cuneiform. The metatarsal engagement member can be positioned in contact with the external surface of the skin of the patient overlaying the medial side of the metatarsal. The tip can be inserted through the small stab incision between the adjacent metatarsals. For example, the tip can be inserted through the small stab incision and the tip positioned in contact with a lateral face of a metatarsal different than the metatarsal being moved. In either case, the bone positioner may include one more mechanisms that allow the metatarsal engagement member to be moved relative to the tip, resulting in the metatarsal moving in one or more planes relative to the cuneiform.

[0047] With the metatarsal suitably repositioned, the clinician may temporarily and/or permanently fixate the moved position of the metatarsal relative to the cuneiform. For example, the clinician may insert a fixation pin through the metatarsal and into the cuneiform (e.g., across the TMT joint) to hold the position of the metatarsal for subsequent installation of one or more permanent fixation devices. In some implementations, the clinician compresses the prepared end face of the metatarsal together with the prepared end face of the cuneiform before installing one or more temporary and/or permanent fixation devices.

[0048] In either case, in some examples, the clinician may grasp a plate holder instrument operatively connected to a bone plate and use the instrument to position the bone plate through the comparatively small incision. Through manipulation of the plate holder instrument, the clinician may position and hold the bone plate across the TMT joint while the clinician inserts one or more fixation elements through one or more corresponding fixation holes in the bone plate into the underlying metatarsal and/or cuneiform.

[0049] FIG. **4** is a flow diagram illustrating an example technique for performing a minimally invasive metatarsal realignment procedure. The example technique will be described with respect to first metatarsal **210** and medial cuneiform **222**, although can be performed on other bones, as discussed above. For purposes of discussion, the technique of FIG. **4** will be discussed with respect to different example images illustrated in FIGS. **5-22**.

[0050] FIG. **5** is a perspective view of a foot **200** showing example surgical access to the TMT joint. With reference to FIGS. **4** and **5**, the example technique involves surgically accessing TMT joint **230** separating first metatarsal **210** from opposed medial cuneiform **222** (**300**). To surgically access the joint, the patient may be placed in a supine position on the operating room table and general anesthesia or monitored anesthesia care administered. Hemostasis can be obtained by applying thigh tourniquet or mid-calf tourniquet. An incision **400** (FIG. **5**) can be made through the skin **402**, such as on a dorsal side of the foot, a medial side of the foot, or on a dorsal-medial side of the foot.

[0051] To help identify the proper location to make incision **400**, in some implementations, the clinician may image foot **200** prior to making the incision. For example, the clinician may take a fluoroscopic (e.g. X-ray) image of at least a portion of foot **200** encompassing TMT joint **230** prior to making incision **400**. Imaging foot **200** can reveal to the clinician the location of TMT joint **230** about which incision **400** can be centered when subsequently cutting through skin **402**. In these applications, the clinician may use a tool such as a ruler, incision guide, or other instrument fabricated at least partially from a radiopaque material to designate the location of TMT joint **230** under imaging.

[0052] FIG. **6** is an image of one example configuration of an incision guide **404** that can be used to help designate the location of TMT joint **230** prior to making incision **400**. In this example, incision guide **404** is shown as including one or more radio-identifiable marking lines **406**, which are distinguishable from a remainder of the incision guide under imaging. The one or more radio-identifiable marking lines **406** can be formed from a different material than a remainder of the incision guide, have a different thickness than a remainder of the incision guide, and/or otherwise be distinguishable under imaging from the remainder of the incision guide. In either configuration, the clinician may align radio-identifiable marking line **406** with TMT joint **230** under imaging to designate the location for subsequently making incision **400**.

[0053] FIG. **7** is a perspective view of foot **200** showing incision guide **404** positioned over metatarsal **210** with radio-identifiable marking line **406** generally co-planar with TMT joint **230** between the metatarsal and medial cuneiform **222**. By accurately identifying the location of TMT joint **230** prior to making incision **400**, the clinician may make a comparatively smaller incision substantially centered on the TMT joint than if the clinician makes an exploratory incision without first identifying the precise location of the joint.

[0054] Independent of whether the clinician does or does not utilize an imaging device to identify TMT joint **230** prior to making incision **400**, the clinician can make the incision to access the underlying joint. While the specific length of incision **400** (in the distal to proximal direction parallel to the long axis of first metatarsal **210**) may vary depending on the anatomy of the specific patient undergoing the procedure and the clinician performing the surgery, in some examples, the incision may have a length less than 12 cm. For example, the clinician may cut incision **400** to have a length less than 10.5 cm, less than 8 cm, less than 6 cm, or less than 5 cm. In some instances, incision **400** may have a length ranging from 1 cm to 12 cm, such as from 1.5 cm to 8 cm, 2 cm to 6 centimeters, or from 3 cm to 4 cm.

[0055] With TMT joint **230** exposed, the technique of FIG. **4** involves preparing the end face of first metatarsal **210** and/or preparing the end face of medial cuneiform **222** (**302**, **304**). While FIG. **4** schematically illustrates an example order in which the end face of the first metatarsal and the end face of the medial cuneiform can be prepared, it should be appreciated that the surgical technique is not limited to any particular order of preparation. For example, the clinician can prepare the end face of either the metatarsal or the cuneiform before preparing the end face of the other bone. Further, one or both of the end faces of the metatarsal and the cuneiform can be prepared before and/or after the metatarsal is moved relative to the cuneiform. Accordingly, unless otherwise specified, the order of bone preparation and/or movement is not limited.

[0056] In general, the clinician can prepare the end of each bone forming a TMT joint so as to promote fusion of the bone ends across the TMT joint following realignment. Bone preparation may involve using a tissue removing instrument to apply a force to the end face of the bone so as to create a bleeding bone face to promote subsequent fusion. Example tissue removing instruments that can be used include, but are not limited to, a saw, a rotary bur, a rongeur, a reamer, an osteotome, a curette, and the like. The tissue removing instrument can be applied to the end face of the bone being prepared to remove cartilage and/or bone. For example, the tissue removing instrument may be applied to the end face to remove cartilage (e.g., all cartilage) down to subchondral bone. Additionally or alternatively, the tissue removing instrument may be applied to cut, fenestrate, morselize, and/or otherwise reshape the end face of the bone and/or form a bleeding bone face to promote fusion. In instances where a cutting operation is performed to remove an end portion of a bone, the cutting may be performed freehand or with the aid of a cutting guide having a guide surface positionable over the portion of bone to be cut. When using a bone preparation guide, a cutting instrument can be inserted against the guide surface (e.g., between a slot define between two guide surfaces) to guide the cutting instrument for bone removal.

[0057] In some examples, the clinician cuts at least one bone defining the TMT joint **230** (e.g., one or both of first metatarsal **210** and medial cuneiform **222**). The clinician may cut both bones defining the TMT joint or may cut only one bone defining the joint and perform a different preparation technique on the other bone.

[0058] While a variety of different bone preparation guides can be used to guide a cutting instrument, FIG. **8** illustrates one example bone preparation guide **150** that may be used as part of a minimally invasive procedures. In the illustrated example, bone preparation guide **150** includes a body **154** defining at least one guide surface for guiding a cutting instrument which, in the illustrated example, is shown as including two guide surfaces: a first guide surface **160** to define a first preparing plane and a second guide surface **164** to define a second preparing plane. A tissue removing instrument (e.g., a saw, rotary bur, osteotome, etc.) can be aligned with and/or guided along (e.g., in contact with) the surface(s) to remove tissue (e.g., remove cartilage or bone and/or make cuts to bone). The first and second guide surfaces **160**, **164** can be spaced from each other by a distance, (e.g., between about 2 millimeters and about 10 millimeters, such as between about 4 and about 7 millimeters). In the embodiment shown, the first and second guide surfaces are parallel, such that cuts to adjacent bones using the guide surfaces will be generally parallel. However, in other examples, one guide surface can be angled relative to another guide surface at a non-zero degree angle.

[0059] In some configurations, as shown in FIG. **8**, a first facing surface **166** is positioned adjacent the first guide surface **160** and/or a second facing surface **168** is positioned adjacent the second guide surface **164**. In such configurations, the distance between the first guide surface and the first facing surface defines a first guide slot, and the distance between the second guide surface and the second facing surface defines a second guide slot. Each slot can be sized to receive a tissue removing instrument to prepare the bone ends. The first and second slots may be parallel or skewed. In the illustrated example, the facing surfaces each contain a gap, such that the surface is not a single, continuous surface. In other embodiments, one or both facing surfaces can be a single, continuous surface lacking any such gap.

[0060] The space between adjacent guide surfaces and/or cutting slots (when configured with multiple of such features) can be solid material or can include one or more openings. In the illustrated example, an opening **170** is defined by body **154** between the first and second guide surfaces. The opening can be an area between the guide surfaces useful for allowing a practitioner to have a visual path to bones during bone preparation and/or to receive an instrument. In the configuration shown, the opening extends across the body and a distance from a surface **172** opposite of the first facing surface **166** to a surface **174** opposite of the second facing surface **168**.

[0061] The illustrated bone preparation guide also includes a first end **176** extending from the body **154** in a first direction and a second end **178** extending from the body in a second direction. The second direction can be different than the first direction (e.g., an opposite direction). As shown, each of the first end and the second end can include at least one fixation aperture **180** configured to receive a fixation pin to secure the bone preparation guide to an underlying bone. For example, first end **176** of bone preparation guide **150** may define a first fixation aperture through which a first pin can be inserted and the second end **178** of bone preparation guide **150** may define a second fixation aperture through which a second pin can be inserted. These two fixation apertures may be parallel aligned, such that first and second pins **112**, **114** extend through the holes parallel to each other. The first end **176** and/or the second end **178** of bone preparation guide **150** may also define one or more additional fixation apertures that are angled (at a non-zero degree angle) or otherwise skewed relative to the two parallel fixation apertures.

[0062] In use, a clinician may insert the two parallel pins through fixation apertures **180** and may optionally insert one or more angled pins through the one or more angled fixation apertures. This combination of parallel and angled pins may prevent bone preparation guide **150** from being removed from the underlying bones being worked upon. When the clinician has completed using the bone preparation guide, the angled pin or pins may be removed leaving the two parallel pins inserted into the underlying bones. Bone preparation guide **150** can be slid or otherwise moved up and off the parallel pins and, in some examples, a compressor (e.g., a compressor-distractor) thereafter inserted down over the pins. The compressor can then be used to apply a force to the pins to compress the prepared ends of the two facing bones together.

[0063] In some examples as shown in FIG. **8**, bone preparation guide **150** can also include a first adjustable stabilization member **182** engaged with the first end **176** and/or a second adjustable stabilization member **184** engaged with the second end **178**. Each of the members can be threaded and engage a threaded aperture defined by the ends. The elevation of each end can be adjusted with respect to a bone by adjusting the stabilization member. In some embodiments, as shown, the stabilization members are cannulated such that they can receive a fixation pin. In other examples, bone preparation guide **150** does not include stabilization members and/or has a different configuration than that specifically illustrated. Details on example bone realignment instruments and techniques that can be used in conjunction with the present disclosure are described in U.S. Pat. No. 9,622,805, issued Apr. 18, 2017 and entitled "BONE POSITIONING AND PREPARING GUIDE SYSTEMS AND METHODS" the entire contents of which are incorporated herein by reference.

[0064] With reference to FIG. **9**, bone preparation guide **150** may include or be used with a spacer **188** that extends downward from the body **154**. Spacer **188** may be configured to be placed into a joint (e.g., within the TMT joint). In some embodiments, the spacer **188** is selectively engageable with the body of the bone preparation guide and removable therefrom. The spacer can have a first portion **190** configured to extend into a joint space and a second portion **192** engageable with the body **154**. In the embodiment shown, the spacer can be received within opening **170**, such that the spacer extends from the body in between the first and second guide surfaces. Such a spacer can be useful for positioning the body at a desired position with respect to a joint and for properly positioning the guide with respect to bones to be cut. The distance between the spacer and the first guide surface can define a length of tissue removal (e.g., bone or cartilage to be cut) from a first bone, and the distance between the spacer and the second guide surface can define a length of tissue removal (e.g., bone or cartilage to be cut) from a second bone.

[0065] When configured with spacer **188**, the spacer can be inserted into TMT joint **230** after making incision **400**. After inserting spacer **188** into the TMT joint, opening **170** of bone preparation guide **150** can be aligned with the spacer and the bone preparation guide installed one the spacer, thereby positioning the one or more guide surfaces defined by the bone preparation guide over one or more bone ends to be prepared. In other examples, spacer **188** may be engaged with bone preparation guide **150** prior to installation in TMT joint **230** and the spacer thereafter inserted into the joint as an assembly connected to the bone preparation guide. In still other examples, spacer **188** may be integrally and permanently attached to bone preparation guide **150** or the bone preparation guide may not include a spacer.

[0066] As also shown in FIG. **9**, bone preparation guide **150** may include or be used with a tissue removal location check member **194**. Tissue removal check member **194** may be engageable with the body **154** and configured to extend to a first bone and a second bone. The tissue removal location check member can have a first portion **196** configured to extend into contact with first and second bones and a second portion **198** engageable with the body. In the embodiment shown, the tissue removal check member **194** is configured to extend in the body **154** at both the first and second guiding surfaces. The tissue removal location check member **194** may be useful for allowing a practitioner to see where a tissue removing instrument guided by the surfaces will contact the bone to be prepared.

[0067] In addition to or in lieu of using a spacer, the systems and techniques according to the disclosure may utilize a fulcrum. A fulcrum may be a device positionable between the proximal base of the metatarsal being moved (e.g., first metatarsal **210**) and an adjacent metatarsal (e.g., second metatarsal **212**). The fulcrum can establish and/or maintain space between adjacent bones being moved, preventing lateral translation or base shift of the bones during rotation and/or pivoting.

[0068] When used, the fulcrum may be a standalone instrument or may be coupled to spacer **188** to define a bi-planar instrument. A bi-planar instrument can define a spacer body coupled to a fulcrum body that may be useful to provide a unitary structure (e.g., prior to or after being assembled) that can be positioned between two adjacent, intersecting joint spaces: a first joint space between opposed ends of a metatarsal and cuneiform and an intermetatarsal space between adjacent metatarsals. The spacer body can include a first portion insertable into the joint space and a second portion that projects above the joint space. The second portion projecting above the joint space can be coupled to a bone preparation guide, thereby facilitating positioning of the bone preparation guide over the metatarsal and/or cuneiform between which the spacer body is positioned. The fulcrum body can establish and/or maintain space between adjacent bones being moved, preventing lateral translation or base shift of the bones during rotation and/or pivoting.

[0069] FIG. **10** is a perspective view of an example foot **200** in which a bi-planar instrument is positioned in a first joint space and an intersecting second joint space, where a bone forming the first and second joint spaces is being realigned relative to one or more adjacent bones. In particular, FIG. **10** illustrates a bi-planar instrument having a spacer body **188** coupled to a fulcrum body **189**. Spacer body **188** is positioned at an intersection between an end of first metatarsal **210** and medial cuneiform **222**. Fulcrum body **189** is positioned between first metatarsal **210** and second metatarsal **212**.

[0070] With further reference to FIG. **4**, either before or after preparing one or both ends of first metatarsal **210** and medial cuneiform **222**, the clinician may move the metatarsal in at least one plane (**306**). For example, the clinician may move metatarsal **210** in at least the transverse plane to close an intermetatarsal angle between the metatarsal and an adjacent bone (e.g., a second metatarsal) and/or a frontal plane (e.g., to reposition the sesamoid bones substantially centered under the metatarsal). In some examples, the clinician moves the bone portion in multiple planes, such as the transverse plane and/or frontal plane and/or sagittal plane. The clinician may or may not utilize a bone positioning device to facilitate movement of the bone portion.

[0071] FIG. **11** shows a side perspective view of an example bone positioner **10** (also referred to as a bone positioning device) that can be used to move a metatarsal relative to an adjacent bone. In some implementations, the bone positioning device includes a metatarsal engagement member, a tip, and a mechanism to move the metatarsal engagement member and the tip relative to each other in one or more planes. For example, the mechanism may move the metatarsal engagement member and the tip towards each other (e.g. moving the metatarsal engagement member towards the tip, moving the tip towards the metatarsal engagement member, or moving both simultaneously). The bone positioning device may also include an actuator to actuate the mechanism. When the mechanism engaged, it can cause the metatarsal engaged with the metatarsal engagement member to move to correct an alignment in at least one plane with respect to a second bone in contact with the tip.

[0072] In the embodiment of FIG. **11**, bone positioning device **10** includes a main body member **20**, a shaft **30**, a metatarsal engagement member **40** is connected to the shaft, and a tip **50** is connected to the main body member. In general, main body member **20** can be sized and shaped to clear anatomy or other instrumentation (e.g., pins and guides) while positioned on a patient. In the embodiment of FIG. **11**, the main body member **20** includes a generally C-shaped. In some embodiments, the main body is sized and configured to engage bones of a human foot. In addition, although bone positioning device **10** is illustrated as being formed of two components, main body member **20** and shaft **30**, the guide can be fabricated from more components (e.g., three, four, or more) that are joined together to form the guide.

[0073] A shaft **30** can be movably connected to the main body member **20**. In some embodiments, the shaft **30** includes threads **80** that engage with the main body member **20** such that rotation of the shaft translates the shaft with respect to the main body member. In other embodiments, the shaft can slide within the main body member and can be secured thereto at a desired location with a set screw. In yet other embodiments, the shaft can be moved with respect to the main body by a ratchet mechanism or yet other mechanism that rotates and/or linearly translates metatarsal engagement member **40** relative to tip **50**. In the embodiment shown, the shaft moves along an axis that intersects the tip **50**. In other embodiments, the shaft **30** and/or metatarsal engagement member **40** is offset from tip **50**.

[0074] In general, metatarsal engagement member **40** is configured (e.g., sized and/or shaped) to be positioned in contact with an external surface of the skin **402** of the patient. As a result, force applied to and/or through metatarsal engagement member **40** is directed through the skin to the underlying metatarsal **210** rather than directly to the metatarsal. As a result, metatarsal engagement member **40** may be positioned against the skin of the patient without making an incision to facilitate direct contact between the metatarsal engagement member **40** and metatarsal **210**.

[0075] FIG. **12** is an illustration showing an example arrangement of bone positioning device **10** in which metatarsal engagement member **40** is positioned in contact with the external surface of skin **402** of the patient covering a side of metatarsal **210**. Bone positioning device **10** can apply a force to move metatarsal **210** through skin **402** in one or more planes. In the illustrated configuration, metatarsal engagement member **40** is positioned in contact with the external surface of skin **402** of the patient covering a medial side of metatarsal **210**. For example, metatarsal engagement member **40** may be positioned in contact with a portion of skin that covers the medial-most half of metatarsal **210**. This arrangement is useful, e.g., to move metatarsal **210** laterally to close the IMA between first metatarsal **210** and second metatarsal **212**. In other configurations, the metatarsal engagement member **40** may be positioned in contact with the external surface of skin **402** of the patient covering a different portion of metatarsal **210**. In still further implementations, the techniques of the disclosure may be performed without a bone positioning device or a bone positioning device that contacts metatarsal **210** directly rather than through skin **402**. However, configuring bone positioning device **10** to deliver movement force through skin **402** can help facilitate a minimal incision procedures.

[0076] With further reference to FIG. **11**, metatarsal engagement member **40** may be configured (e.g., sized and/or shaped) to contact skin overlying an underlying bone (e.g., first metatarsal **210**). Metatarsal engagement member **40** may define a concave shape to generally conform and/or wrap partially around the underling cylindrical bone. The concave shape may include define a continuous radius of curvature, a V-shape, a planer region between outwardly extending sidewalls, and/or other shape having a concavity. In still other examples, metatarsal engagement member **40** may be planar. In either case, when configured with a concave shape, the bottom of the concavity can be centered between outwardly extending sidewalls **42**, **44**. Each sidewall can extend outward to the same height and/or at the same slope, or may extend to different heights and/or at different slopes.

[0077] While the specific dimensions of metatarsal engagement member **40** may vary depending on the manufacturer and/or user preferences, in some examples, the metatarsal engagement member may have a length sufficient at least partially conform around the skin-covered surfaces of the metatarsal being moved (e.g., first metatarsal **210**). For example, metatarsal engagement member **40** may have a length **46** extending in a dorsal to plantar direction (when positioned in contact with a patient) that is at least 20 mm, such as at least 25 mm, at least 28 mm, at least 30 mm, or at least 35 mm. For example, the length **46** may range from 28 mm to 35 mm.

[0078] To help position bone positioning device **10** in the sagittal plane when installing on a patient's foot, metatarsal engagement member **40** may include a center line indicator **48**. The center line indicator **48** may be a visual indicator (e.g., line) indicating a geometric center of metatarsal engagement member **40** between the dorsal and plantar ends of the metatarsal engagement member. In use, the clinician may substantially align center line indicator **48** with a midpoint (e.g., geometric center between the dorsal and plantar surfaces) of the metatarsal being moved (with the midpoint being covered by skin **402**). The clinician may identify the midpoint by visual and/or tactile inspection and/or through radiographic (e.g., fluoroscopic) imaging. Additionally or alternative, the clinician can mark (with a surgical pen) the midpoint on the skin as a reference for alignment. Alignment of the metatarsal engagement member can then occur by aligning the center indicator mark to the marked midpoint of the metatarsal.

[0079] In the embodiment shown, the metatarsal engagement member **40** is provided on an end of the shaft **30**. In the embodiment of the shaft shown having threads **80**, the metatarsal engagement member **40** can be rotatably coupled to the shaft **30**. In such embodiments, as the shaft is rotated relative to the main body member the metatarsal engagement member **40** may or may not rotate with respect to the main body member even as it translates with respect to the main body member along with the shaft **30** and rotates with respect to the shaft. The metatarsal engagement member may oscillate about the shaft **30**, but generally does not rotate with respect to skin **402** and/or underlying bone after contact with the skin.

[0080] Tip **50** can be useful for contacting a bone, such as a bone different than the bone being moved by bone positioning device **10**. For example, if metatarsal engagement member **40** is in contact with skin covering first metatarsal **210**, the tip can be in contact with a lateral side of a different metatarsal (e.g., the second, third, fourth, or fifth metatarsal) and/or a lateral side of skin covering such metatarsal. In some examples, a small stab incision may be made between the different metatarsal and a laterally-adjacent metatarsal. For example, a stab incision may be made in an intermetatarsal space between the second metatarsal **212** and third metatarsal **214**. Tip **50** of bone positioning device **10** can be inserted through the incision and the tip positioned in contact with the lateral side of the second metatarsal.

[0081] In different configurations, tip **50** may be straight or may be tapered to facilitate percutaneous insertion and contact with bone. The tip can also include a textured surface, such as serrated, roughened, cross-hatched, knurled, etc., to reduce slippage between the tip and bone. In the embodiment shown, tip **50** further includes a stop **52**. Depth stop **52** can limit a depth of insertion into an intermetatarsal space (e.g., by contacting a dorsal surface of the metatarsal against which tip **50** is intended to be positioned).

[0082] As shown in FIG. **11**, bone positioning device **10** can also include an actuator (e.g., a knob or a handle) **60** to actuate the mechanism, in this embodiment associated with the shaft. In the embodiment shown, the actuator can be useful for allowing a user to rotate the shaft with respect to the main body member **20**. Actuator **60**, shaft **30**, and/or metatarsal engagement member **40** may include a cannulation **70** extending therethrough to allow the placement of a fixation wire (e.g., K-wire) through these components and into contact with or through a bone engaged with the metatarsal engagement member. For example, a fixation wire can be placed into the bone engaged with metatarsal engagement member **40** to fix the position of the metatarsal engagement member with respect to the bone. In another example, the fixation wire can be placed through the bone in contact with the metatarsal engagement member and into an adjacent bone to maintain a bone position of the bone in contact with the metatarsal engagement member and the adjacent bone.

[0083] Embodiments of the bone positioning device may include any suitable materials. In certain embodiments, the bone positioning device is fabricated at least partially from a radiolucent material such that it is relatively penetrable by X-rays and other forms of radiation, such as thermoplastics and carbon-fiber materials. Such materials are useful for not obstructing visualization of bones using an imaging device when the bone positioning device is positioned on bones.

[0084] FIG. **13** is a perspective view of foot **200** showing seeker **188** and fulcrum **189** inserted between adjacent joint spaces and bone positioner **10** engaged with first metatarsal **210**. FIG. **14** is a perspective view of foot **200** from FIG. **13** further illustrated bone preparation guide **150** installed on seeker **188**. In different examples, bone positioner **10** may be installed before and/or after installing bone preparation guide **150** and/or before or after preparing one or both bone ends. FIG. **15** is a further illustration of the surgical instrument arrangement of FIG. **14** showing the example location of incision **400** and skin **402** relative to the surgical instruments.

[0085] Independent of whether the clinician utilizes a bone positioning device or the configuration of such guide, the clinician can move metatarsal **210** in one or more planes. The clinician may move the metatarsal to help correct an anatomical misalignment of the metatarsal. For example, the clinician may move the metatarsal so the metatarsal is anatomically aligned in one or more planes (e.g., two planes, three planes).

[0086] In some examples, the clinician manually moves first metatarsal **210** in addition to or moving the metatarsal through a force applied by bone positioner **10**. For example, the clinician may grasp a pin inserted into the first metatarsal and/or grasp the metatarsal with a tool (e.g., forceps) and manipulate the position of the metatarsal to a desired position. In some applications, bone positioner **10** may apply a force the moves first metatarsal **210** in the transverse plane to close the IMA with an adjacent metatarsal. However, the amount of frontal plane rotation and/or sagittal plane movement achieved by bone positioner **10** through the skin of the patient may be limited. Accordingly, the clinician may manually move first metatarsal **210** (e.g., by grasping a pin inserted into the metatarsal and at least partially projecting out of the metatarsal) to rotate the metatarsal in the frontal plane and/or move the metatarsal in the sagittal plane.

[0087] In general, an anatomically aligned position means that an angle of a long axis of a first metatarsal relative to a long axis of a second metatarsal is about 10 degrees or less in the transverse plane or sagittal plane. In certain embodiments, anatomical misalignment can be corrected in both the transverse plane and the frontal plane. In the transverse plane, a normal intermetatarsal angle ("IMA") between a first metatarsal and a second metatarsal is less than about 9 degrees. An IMA of between about 9 degrees and about 13 degrees is considered a mild misalignment of the first metatarsal and the second metatarsal. An IMA of greater than about 16 degrees is considered a severe misalignment of the first metatarsal and the second metatarsal. In some embodiments, the first metatarsal is moved to reduce the IMA from over 10 degrees to about 10 degrees or less (e.g., to an IMA of about 1-5 degrees), including to negative angles of about −5 degrees or until interference with the second metatarsal, by positioning the first metatarsal at a different angle with respect to the second metatarsal.

[0088] With respect to the frontal plane, a normal first metatarsal will be positioned such that its crista prominence is generally perpendicular to the ground and/or its sesamoid bones are generally parallel to the ground and positioned under the metatarsal. This position can be defined as a metatarsal rotation of 0 degrees. In a misaligned first metatarsal, the metatarsal is axially rotated between about 4 degrees to about 30 degrees or more. In some embodiments, methods in accordance with the invention are capable of anatomically aligning the metatarsal by reducing the metatarsal rotation from about 4 degrees or more to less than 4 degrees (e.g., to about 0 to 2 degrees) by rotating the metatarsal with respect to the medial cuneiform.

[0089] In some applications, independent of whether the clinician performs the specific bone realignment technique discussed above, the clinician may compress the end faces of the prepared bones (e.g., first metatarsal **210**, medial cuneiform **222**) together. For example, the technique of FIG. **4** includes optionally compressing the prepared end faces of the bone portions together prior to fixating (**308**). The clinician may compress the end faces together with hand pressure and/or using a compressing instrument physically attached to both the first bone portion and the second bone portion. For instance, the clinician may attach a compressing instrument to metatarsal **210** with one or more fixation pins and also attach the compressing instrument to medial cuneiform **222** using one or more fixation pins. In some examples, the clinician lifts bone preparation guide off two or more pins that extend parallel to each other, at least one of which is inserted into metatarsal **210** and at least one of which is inserted into medial cuneiform **222**. The clinician can then install a compressor back down on the parallel pins.

[0090] When used, the compressing instrument may have a mechanism (e.g., threaded rod, rack and pinion) that presses against the pins inserted through the two bone portions to compress the end faces of the two bone portions together. Additional details on example compressing instruments that may be used can be found in US Patent Publication No. 2020/0015856, published Jan. 16, 2020 and entitled "COMPRESSOR-DISTRACTOR FOR ANGULARLY REALIGNING BONE PORTIONS," the entire contents of which are incorporated herein by reference.

[0091] FIG. **16** is a perspective view of foot **200** illustrating an example compressing instrument **420** that may be used to compress the prepared end faces of first metatarsal **210** and medial cuneiform **222** together. Compressing instrument **420** may be attached using at least one fixation pin inserted through an arm of the compressing instrument into first metatarsal **210** and using at least one fixation pin inserted through an arm of the compressing instrument into medial cuneiform **222**.

[0092] In addition to or in lieu of compressing the prepared end faces of the bones together, the technique of FIG. **4** may involve temporarily or provisionally fixating a moved position of first metatarsal **210** relative to medial cuneiform **220** prior to attaching a permanent fixation device (**310**). In some implementations, the clinician inserts one or more fixation pins through first metatarsal **210** and into an adjacent bone (e.g., second metatarsal **212**, medial cuneiform **222**), such as through the end faces of the first metatarsal **210** and medial cuneiform **220** in addition to or in lieu of compressing the end faces with a compressing instrument. The fixation pin may be a k-wire, olive wire (e.g., pin with region of enlarged cross-section) or other fixation pin structure. The fixation pin crossing joint **230** between first metatarsal **210** and medial cuneiform **220** can help provisionally fixate and/or compress the end faces of the two bone portions together prior to installation of permeant fixation device (and subsequent fusion of the bone faces together). When used, the one or more fixation pins can be removed from the end faces of the two bone portions and across the joint between the bone portions after installation of permanent fixation device.

[0093] Independent of whether the clinician provisionally fixates the moved position of first metatarsal **210** relative to medial cuneiform **220**, the technique of FIG. **4** involves installing one or more permanent fixation devices to first metatarsal **210** and medial cuneiform **220** across joint **230** (**312**). The one or more permanent fixation devices may hold the moved position of first metatarsal **210** relative to medial cuneiform **220**, allowing the end faces of the bone portions to fuse together through a subsequent healing process.

[0094] Any one or more bone fixation devices can be used including, but not limited to, a bone screw (e.g., a compressing bone screw), a bone plate, a bone staple, an external fixator, a pin (e.g., an intramedullary implant), and/or combinations thereof. The bone fixation device may be secured on one side to metatarsal **210**, bridge the TMT joint **230**, and be secured on an opposite side to the proximal medial cuneiform **222**

[0095] In one example, two bone plates may be placed across the TMT joint to provide bi-planar plating. For example, a first bone plate may be positioned on a dorsal side of the metatarsal and medial cuneiform. A second bone plate may be positioned on a medial/dorsal-medial side of the metatarsal and medial cuneiform. Independent of the number or configuration of bone plates, the plates may be applied with the insertion of bone screws.

[0096] A variety of different of different bone plate configurations can be used to permanently fixate a metatarsal to a cuneiform. In some examples, the bone plate is configured as a linear bone plate in which all the fixation holes extending through the bone plate are co-linear with each other. In other examples, the bone plate includes one or more fixation holes extending through one or more branch(es) that are non-co-linear with the remaining co-axially arranged fixation holes. For instance, in various examples, the bone plate may include at least one branch extending outwardly from a longitudinal axis of the bone plate to define at least one of a Y-shape, an L-shape, a T-shape, a U-shape, and/or other shape profile.

[0097] FIG. **17** is illustration of an example bone plate **100** that can be used to help permanently fixate a moved position of a metatarsal relative to an opposed cuneiform. In particular, bone plate **100** illustrates an example bone plate having a U-shaped profile. Configuring bone plate **100** with a U-shaped profile can be useful to help facilitate a minimally invasive surgical procedure. Because of the comparatively short length of the surgical incision in such a procedure, installation of a straight fixation plate through the incision may result in one or more fixation holes being positioned under the skin of the patient and therefore inaccessible for the clinician to insert screws through the fixation holes. By configuring bone plate **100** with a shaped profile (e.g., U-shape), one or more fixation holes extending through the arm(s) of the U-shape may wrap into a region of the incision accessible by the clinician for inserting screws.

[0098] As shown in FIG. **17**, bone plate **100** can have a body **102** defining a base **104**, a first arm

106 extending at an angle from the base, and a second arm **108** extending at an angle from the base. The first and second arms **106**, **108** may extend at the same angle away from base **104** or may extend at different angles. In some examples, the first and second arms **106**, **108** each extend at an angle **110** from base **104** ranging from 30 degrees to 175 degrees, such as from 45 degrees to 150 degrees, or from 75 degrees to 135 degrees. For example, angle **110** may be greater than or equal to 90 degrees.

[0099] In either case, bone plate **100** may include one or more fixation holes extending through the thickness of body **102**. In these examples, body **102** may include one or more fixation holes in first arm **106**, one or more additional fixation holes in second arm **108**, and base **104** may or may not include any fixation holes.

[0100] In the illustrated example, first arm **106** has at least one fixation hole, which is illustrated as two fixation holes **112** and **114**. Second arm **108** also has at least one fixation hole, which is also illustrated as two fixation holes **116** and **118**. Fixation hole **114** is at the intersection of first arm **106** and base **104**. Fixation hole **118** is at the intersection of second arm **108** and base **104**. Base includes a region between fixation holes **114** and **118** the forms a bridge devoid of any fixation holes, which can be placed bridging across TMT joint **230**.

[0101] In the illustrated configuration, one or more drill guides **120** are threadingly inserted into one or more of the fixation holes. Drill guides **120** can provide a structure for guiding a drill to drill into the underlying bone (e.g., prior to inserting a screw). In some implementations, drill guides **120** may also be used to grasp plate **100** with a plate holder to manipulate the plate during placement.

[0102] Although bone plate **100** is illustrated as being configured with four fixation holes the bone plate may include fewer fixation holes or more fixation holes. For example, first arm **106** and second arm **108** of bone plate **100** may each include fewer fixation holes (e.g., one) and/or more fixation holes (e.g., three, four). The dimensions (e.g., length) of the arms and/or base can be adjusted to accommodate the particular number of fixation holes, among other factors.

[0103] Each feature described as a fixation hole may be an opening extending through the thickness of body **102** that is sized to receive a corresponding screw. Each fixation hole may typically have a circular cross-sectional shape although may have other cross-sectional shapes without departing from the scope of the disclosure. The fixation hole may extend perpendicularly through the thickness of body **102** or may taper (e.g., such that the hole is larger adjacent the top surface of the plate than adjacent the bone-facing surface of the plate). In still other examples, the fixation hole may extend at a non-perpendicular angle through the thickness of body **102** and/or be configured for polyaxial screw alignment. In either case, bone plate **100** can be secured to underlying bone portions by inserting fixation elements through corresponding fixation holes. Example fixation elements that can be used are screws, may be locking screws and/or non-locking screws.

[0104] When configured with a U-shape, bone plate **100** may define a maximum length **122** less than the combined length of the individual linear segments of the plate (e.g., the combine lengths **124**, **126**, and **128**). In some examples, maximum length **112** is less than the length of incision **400** whereas the combined length of the individual linear segments of the plate may have a length greater than the length of incision **400**. As a result, fixation holes (e.g., **112**, **116**) that be positioned at a location accessible through incision **400** when bone plate **100** is configured with a U-shape but may otherwise be inaccessible if the same place is reconfigured as a linear bone plate.

[0105] Access may be particularly challenging when the clinician positions the bone plate at a location offset from the incision location. For example, the clinician may position the base of the U-shaped bone plate at location that is offset at least 45 degrees from the center of the incision, such as at least 60 degrees, at least 75 degrees, or approximately 90 degrees or more. For instance, the clinician may make a dorsal incision and position the base of the U-shaped bone plate on a medial side of the TMT joint. The clinician may position the base of the bone plate across the TMT joint with the legs extending upwardly in a dorsal direction relative to the base.

[0106] FIG. **18** is a perspective view of foot **200** showing an example insertion location of bone plate **100**. As shown in this example, the arms of U-shaped bone plate wrap into region accessible by through incision **400**. In this example, a second bone plate **140** that has a linear profile is attached on a dorsal side of the metatarsal and cuneiform, across the TMT joint.

[0107] To install bone plate **100** (or another bone plate) in a minimal incision procedure, the clinician may utilize a plate holder to help manipulate and/or hold the bone plate within the incision. The plate holder may be an instrument configured to releasably grasp an external surface of the plate and/or be inserted into a fixation hole of the plate, such as a drill guide inserted through the fixation hole. Using the plate holder, the clinician may be able to controllably and precisely position the bone plate and/or hold the bone plate during installation of one or more screws in the small incision created to access the TMT joint space. This can help address space constrains caused by the comparatively small length incision, which can make it difficult for the clinician to properly position the bone plate using their fingers along.

[0108] In general, a plate holder may be any instrument that can be releasably connected to a bone plate to help facilitate installation of the bone plate and then detached after and/or during installation. In various examples, the plate holder may be operatively coupled to an exterior surface of the bone plate (e.g., by grasping a protrusion from the plate and/or opposed sides of the plate) and/or an interior surface of the bone plate (e.g., by inserting into one or more fixation holes of the bone plate). Further purposes of installation, when one or more drill guides **120** are inserted into the fixation holes of the bone plate, the drill guides may be considered part of the bone plate that the plate holder can grasp and/or be inserted into.

[0109] FIG. **19** illustrates one example plate holder **450** that is configured as a tenaculum or forceps-style plate holder. In this example, plate holder **450** includes a first arm **452** and a second arm **454** that are pivotably connected together. Each arm extends from a proximal end **456** to a distal end **458**. The distal end of each arm may be linear or, as illustrated, may be curved to define opposed converging radiuses of curvature. When so configured, the arms may be engaged with (e.g., wrapped at least partially about) a feature of bone plate **100**. As illustrated, arms **452**, **454** are wrapped around drill guides **120** inserted into fixation holes of the bone plate. In other examples, the bone plate may include a feature (e.g., a recess, cavity, projection) that one or both arms **452**, **454** can be operatively engaged with (e.g., when the tenaculum is closed) to releasably secure the plate holder to the bone plate. In some examples, plate holder **450** includes a locking mechanism (e.g., ratchet, set screw, etc.) that can lock the arms **452**, **454** at a fixed position (e.g., when grasping plate **100**) but be released by the clinician to remove the plate holder from the plate.

[0110] FIG. **20** illustrates another example plate holder **470** configured to releasably attached to one or more drill guides **120** inserted into one or more fixation holes of a bone plate, such as bone plate **100**. Plate holder **470** may include a handle **472** connected to a body defining one more slots and/or apertures **474** into which a drill guide **120** can be inserted to releasably attach the bone plate to the plate holder.

[0111] FIG. **21** illustrates another example plate holder **480** that is configured as a plate bender-style plate holder. In this example, plate holder **480** defines a shaft **482** terminating in a distal end **484**. In different configurations, the distal end **484** of shaft **482** can be configured (e.g., sized and/or shaped) to be inserted into a fixation hole of a bone plate and/or drill guide **120** inserted into the fixation hole and/or to be inserted over an exterior surface of the drill guide (as illustrated in FIG. **21**). In some examples, plate holder **480** may include a retention mechanism **486**, such as a spring, clasp, O-ring, or other device to provide frictional engagement between the plate holder and bone plate (e.g., drill guide inserted into the bone plate) to help releasably attached the plate holder to the bone plate. In use, the clinician may engage one or more plate holders **480**, such as at least two plate holders positioned on different side of the bone plate bridge that crosses the TMT joint, and use the plate holders to manipulate the position of the plate. Additionally or alternatively, the clinician may use the plate holders to bend the bone plate to a desired shape, e.g., specific to the anatomy of the patient undergoing the procedure. In some examples, incision guide **404** also functions as plate holder **480** and/or a plate bender.

[0112] Independent of the specific configuration of a plate holder may be operatively connected to a bone plate. The clinician may manipulate the plate holder and, correspondingly, bone plate, to insert the bone plate through incision **400** until the bone plate is positioned across the TMT joint with at least one fixation hole positioned over an underlying metatarsal and at least one fixation hole positioned over an underlying cuneiform. The clinician can then optionally drill a hole into each underlying bone portion through a bone plate fixation hole (e.g., via a drill guide), remove the drill guide, and then insert a screw to secure the bone plate to the underlying bone. In some examples, the clinician installs screws through fixation holes of the bone plate not attached to the plate holder then removes the plate holder and inserts screws through the remaining fixation holes.

[0113] In general, bone plate **100** may be planar (e.g., in the X-Z plane indicated on FIG. **17**) or may include one or more bends or curves that position at least a portion of the bone plate out of plane with a reminder of the bone plate. For example, as illustrated in FIG. **17**, the ends of the legs of the bone plate (e.g., through which fixation hole **112**, **116** are positioned) may be out of plane with the base of the bone plate and/or base **104** may be curved such that fixation holes **114**, **118** are out of plane with each other.

[0114] In some configurations, the curvature of the bone plate may direct drill guides inserted into adjacent pairs of fixation holes to be directed away from one another, e.g., to form a diverging angle relative to each other rather than a converging angle or be parallel to each other. This may be useful to allow the plate holder to be positioned against the outer surface of the diverging drill guide and help prevent the plate holder from inadvertently slipping off the drill guides.

[0115] As an additional feature, the handle and/or arms of the plate holder may be contoured (e.g., curved) out of plane from the proximal end (e.g., where the clinician's hand holds the instrument) to allow for easy placement against a metatarsal. For example, when so configured, the instrument may allow for the bone plate to be placed on the medial side of the metatarsal with the curvature of the instrument generally conforming to the curvature of the anatomy. This conformity can allow for minimal tissue retraction during plate placement.

[0116] FIG. **22** is an illustration of foot **200** showing an example of plate holder **458** manipulating plate **100** for placement across a TMT joint. While example plate holders have generally been described in conjunction with placement of a U-shaped plate, a plate holder may be releasably attached to, and used to position, a plate having a different configuration (e.g., a straight plate) in addition to or in lieu of a U-shaped plate.

[0117] Various examples have been described. These and other examples are within the scope of the following claims.

Claims

1. A method of performing a metatarsal correction procedure, the method comprising: making an incision to access a tarsometatarsal joint between a first metatarsal and a medial cuneiform; using a bone preparation guide to guide a tissue removing instrument to prepare an end of the first metatarsal and an end of the medial cuneiform through the incision; using a bone positioning device to apply a force to a skin covering the first metatarsal to move the first metatarsal a transverse plane; compressing a prepared end of the first metatarsal and a prepared end of the medial cuneiform together using a compressor; and applying one or more fixation devices to fixate a moved position of the first metatarsal.

2. The method of claim 1, further comprising using the bone positioning device engaged with the skin covering the first metatarsal to move the first metatarsal a sagittal plane.

3. The method of claim 1, further comprising inserting a pin into the first metatarsal and using the

pin to move the first metatarsal in a frontal plane.
4. The method of claim 3, wherein using the bone positioning device to apply the force to the skin covering the first metatarsal to move the first metatarsal the transverse plane and using the pin to move the first metatarsal in the frontal plane comprises correcting a bunion deformity.
5. The method of claim 1, wherein the bone positioning device comprises a metatarsal engagement member, wherein using the bone positioning device comprises positioning the metatarsal engagement member in contact with the skin covering the first metatarsal.
6. The method of claim 5, wherein positioning the metatarsal engagement member in contact with the skin covering the first metatarsal comprises positioning the metatarsal engagement member in contact with the skin covering a medial side of the first metatarsal.
7. The method of claim 5, wherein: the metatarsal engagement member is on an end of a threaded shaft; the bone positioning device comprises a knob connected to the threaded shaft; and using the bone positioning device to apply the force to the skin covering the first metatarsal to move the first metatarsal the transverse plane comprises using the knob to advance the threaded shaft relative to a body of the bone positioning device.
8. The method of claim 7, wherein a cannulation is defined through the knob, threaded shaft, and metatarsal engagement member, and further comprising inserting a wire through the cannulation into the first metatarsal.
9. The method of claim 1, wherein the bone preparation guide comprises a slot defined between a first guide surface and a second guide surface, and using the bone preparation guide to guide the tissue removing instrument to prepare the end of the first metatarsal and the end of the medial cuneiform comprises guiding the tissue removing instrument through the slot.
10. The method of claim 9, wherein the tissue removing instrument comprises a burr.
11. The method of claim 1, further comprising inserting a spacer into the tarsometatarsal joint and using the spacer to position the bone preparation guide with respect to the tarsometatarsal joint.
12. The method of claim 11, wherein using the spacer to position the bone preparation guide with respect to the tarsometatarsal joint comprises aligning an opening of the bone preparation guide with the spacer inserted into the tarsometatarsal joint.
13. The method of claim 1, further comprising, prior to using the bone preparation guide to guide the tissue removing instrument to prepare the end of the first metatarsal and the end of the medial cuneiform through the incision, attaching the bone preparation guide to the medial cuneiform with a pin.
14. The method of claim 1, wherein using the bone positioning device to apply the force to the skin covering the first metatarsal to move the first metatarsal the transverse plane comprises using the bone positioning device to apply the force to the skin covering the first metatarsal to move the first metatarsal the transverse plane after using the bone preparation guide to guide the tissue removing instrument to prepare the end of the first metatarsal and the end of the medial cuneiform.
15. The method of claim 1, wherein the bone positioning guide comprises a tip, and further comprising positioning the tip on a lateral side of a second metatarsal separated from the first metatarsal by an intermetatarsal angle.
16. The method of claim 1, wherein making the incision to access the tarsometatarsal joint between the first metatarsal and the medial cuneiform comprises making an incision within a range 3 cm to 4 cm long.
17. The method of claim 1, wherein compressing the prepared end of the first metatarsal and the prepared end of the medial cuneiform together using the compressor comprises compressing the prepared end of the first metatarsal and the prepared end of the medial cuneiform together using the compressor with the compressor attached to the first metatarsal with a first fixation pin and to the medial cuneiform with a second fixation pin.
18. The method of claim 1, wherein compressing the prepared end of the first metatarsal and the prepared end of the medial cuneiform together using the compressor comprises compressing the

prepared end of the first metatarsal and the prepared end of the medial cuneiform together using the compressor with the compressor attached to using a same fixation pin inserted into the medial cuneiform used to attach the bone preparation guide to the medial cuneiform.

19. The method of claim 1, wherein applying one or more fixation devices to fixate the moved position of the first metatarsal comprises inserting a wire into the first metatarsal.

20. The method of claim 1, wherein applying one or more fixation devices to fixate the moved position of the first metatarsal comprises applying one or more fixation devices across the tarsometatarsal joint.

21. The method of claim 1, wherein the one or more fixation devices comprise a bone screw and/or an intramedullary implant.

22. A method of performing a metatarsal correction procedure, the method comprising: making an incision to access a tarsometatarsal joint between a first metatarsal and a medial cuneiform; using a bone preparation guide to guide a tissue removing instrument to prepare an end of the first metatarsal and an end of the medial cuneiform through the incision, wherein the bone preparation guide comprises a slot defined between a first guide surface and a second guide surface, and using the bone preparation guide to guide the tissue removing instrument to prepare the end of the first metatarsal and the end of the medial cuneiform comprises guiding a burr through the slot; positioning a metatarsal engagement member of a bone positioning device in contact with a skin covering the first metatarsal and using the bone positioning device to apply a force to the skin covering the first metatarsal to move the first metatarsal a transverse plane; inserting a pin into the first metatarsal and using the pin to move the first metatarsal in a frontal plane; compressing a prepared end of the first metatarsal and a prepared end of the medial cuneiform together using a compressor; and applying one or more fixation devices to fixate a moved position of the first metatarsal.

23. The method of claim 22, wherein positioning the metatarsal engagement member in contact with the skin covering the first metatarsal comprises positioning the metatarsal engagement member in contact with the skin covering a medial side of the first metatarsal, and further comprising using the bone positioning device engaged with the skin covering the first metatarsal to move the first metatarsal a sagittal plane.

24. The method of claim 22, wherein using the bone positioning device to apply the force to the skin covering the first metatarsal to move the first metatarsal the transverse plane and using the pin to move the first metatarsal in the frontal plane comprises correcting a bunion deformity.

25. The method of claim 22, wherein: the metatarsal engagement member is on an end of a threaded shaft; the bone positioning device comprises a knob connected to the threaded shaft; using the bone positioning device to apply the force to the skin covering the first metatarsal to move the first metatarsal the transverse plane comprises using the knob to advance the threaded shaft relative to a body of the bone positioning device; and a cannulation is defined through the knob, threaded shaft, and metatarsal engagement member, and further comprising inserting a wire through the cannulation into the first metatarsal.

26. The method of claim 22, further comprising: inserting a spacer into the tarsometatarsal joint; and aligning an opening of the bone preparation guide with the spacer inserted into the tarsometatarsal joint to position the bone preparation guide with respect to the tarsometatarsal joint.

27. The method of claim 22, wherein the one or more fixation devices comprise a bone screw and/or an intramedullary implant.

28. The method of claim 22, wherein making the incision to access the tarsometatarsal joint between the first metatarsal and the medial cuneiform comprises making an incision within a range 3 cm to 4 cm long.

29. The method of claim 22, wherein compressing the prepared end of the first metatarsal and the prepared end of the medial cuneiform together using the compressor comprises compressing the prepared end of the first metatarsal and the prepared end of the medial cuneiform together using the compressor with the compressor attached to using a same fixation pin inserted into the medial cuneiform used to attach the bone preparation guide to the medial cuneiform.